

Hybrid Feature Extraction for Accurate Weather Image Classification

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Abstract—Accurate weather condition recognition from images is crucial for applications like autonomous driving, intelligent transportation, and environmental monitoring. This paper presents a hybrid weather image classification method, integrating ResNet50-based deep feature extraction, followed by Principal Component Analysis (PCA) for feature extraction, and Support Vector Machine (SVM) classification. The approach tackles complex visual patterns and limited annotated data through transfer learning and data augmentation. Experiments on a public Kaggle dataset show high classification accuracy across multiple weather categories, demonstrating its effectiveness and generalizability.

Keywords—weather recognition, ResNet50, SVM, PCA

I. INTRODUCTION

Weather conditions [1], [2] profoundly influence daily activities and industrial operations, impacting transportation safety, outdoor event planning, and renewable energy generation. Accurate image-based weather recognition [3], is vital for safety-critical applications, as it affects visibility and environmental perception for both humans and autonomous systems [4].

However, classifying weather phenomena is challenging due to the intricate and variable visual patterns of meteorological events, the ambiguous boundaries between certain weather states (e.g., light snow vs. fog), and the limited availability of annotated training data.

Traditional weather classification methods depend on physical sensors [7] or human observation [8], but these approaches are constrained by hardware requirements, maintenance costs, and subjectivity. Early computer vision techniques relied on handcrafted features such as SIFT, SURF, and HOG, coupled with classical machine learning algorithms like SVM, k-NN, and decision trees [5], [6] often demand extensive feature engineering and lack generalizability.

Recent advancements in deep learning, particularly Convolutional Neural Networks (CNNs), have significantly enhanced image classification performance [9]. Deeper architectures such as ResNet [10] mitigate training challenges through residual connections, enabling more effective feature extraction. Despite these improvements, deep models typically necessitate large labeled datasets, which are often scarce for specialized weather phenomena.

This paper presents a weather classification system employing a hybrid approach that integrates ResNet50 feature extraction with SVM classification, effectively recognizing diverse weather phenomena from images. Our contributions are:

- Development and validation of a hybrid framework utilizing pretrained ResNet50 for feature extraction and SVM for classification, offering an effective weather image classification solution.
- We developed a hyperparameter optimization strategy using grid search cross-validation to fine-tune the SVM classifier, highlighting the impact of optimal parameter selection on classification performance.
- Addressing data scarcity through strategic data augmentation techniques, including random cropping, horizontal flipping, and rotation, significantly enhancing model generalization.

The paper is structured as follows: Section II reviews existing weather classification methodologies. Section III details the datasets used. Section IV outlines our methodology, including model architecture and training protocols. Section V presents experimental results and comparative analysis. Finally, Section VI concludes with key findings and future research directions.

II. RELATED WORKS

Early image-based weather classification methods relied on traditional machine learning with handcrafted features like color histograms [11] or statistical models [12]. While these methods achieved some success, they heavily depended on domain-specific feature engineering and often struggled to capture the complexity and variability of real-world weather images.

With the rise of deep learning, Convolutional Neural Networks (CNNs) have demonstrated remarkable performance in image classification tasks, including weather recognition. Representative architectures such as EfficientNet [13], Inception V3 [14], and VGG [16] have successively improved image classification [15] accuracy by increasing network depth and iduntrocing innovations like residual connections.

TABLE I. THE NUMBER OF DIFFERENT CATEGORY OF IMAGES

Class	Dew	Fogmog	Frost	Lightning	Hail	Snow
precision	698	851	475	377	591	621
Class	Rain	Rainbow	Frime	Sandstorm	Glaze	
precision	526	232	1160	692	639	

Training deep networks typically demands large labeled datasets, which are scarce for specialized domains like weather phenomena. To address this, transfer learning is employed, leveraging models pre-trained on large datasets like ImageNet [17] and fine-tuning them for specific tasks. Additionally, data augmentation techniques, including geometric and photometric transformations, are widely used to expand dataset size and improve model generalization.

Building on these advances, our work leverages the ResNet50 architecture, which offers an effective balance between model depth and computational efficiency. Unlike previous studies that primarily adopted either end-to-end deep learning or traditional machine learning approaches, we propose a hybrid framework that combines deep feature extraction via ResNet50 with PCA and SVM classification. Furthermore, we address the challenge of limited training data by employing extensive data augmentation and transfer learning strategies, aiming to enhance model robustness and generalization for real-world weather classification tasks.

III. DATASETS

A. Dataset Overview

The dataset employed in this study was obtained from the Kaggle platform, a publicly accessible repository. It encompasses eleven distinct meteorological categories: dew, fogsmog, frost, glaze, hail, lightning, rain, rainbow, rime, sandstorm, and snow. Each category contains a variable number of images, totaling 6,862 samples across the dataset. All images are annotated with their corresponding weather phenomenon labels. The original dataset is derived from the Weather Phenomenon Database (WEAPD) [18]. Table I provides a detailed breakdown of the image distribution across categories.

B. Data Preprocessing

To ensure compatibility with deep learning frameworks, standardized preprocessing steps were performed. Images were resized and center-cropped to $224 \times 224 \times 3$ pixels to align with ResNet50's input requirements. Grayscale images were converted to RGB format. During training, data augmentation techniques, including random resized cropping, horizontal flipping, and rotation (up to 10 degrees), were applied to enhance generalization and mitigate overfitting. Images were then normalized using ImageNet's mean and standard deviation values, adhering to transfer learning practices. Figure 1 illustrates these augmentations.



Fig. 1. Visualization of data augmentation techniques applied to a sample from the "hail" category

The dataset was partitioned into training and validation sets using stratified sampling, with an 80-20 split by default, unless a predefined split was provided. Data loading and batching were managed using PyTorch's DataLoader and ImageFolder utilities, ensuring efficient and reproducible preprocessing.

IV. METHODS

A. ResNet50 Feature Extraction

We utilize the ResNet50 architecture as a deep feature extractor, leveraging its established effectiveness and computational efficiency. Specifically, a pre-trained ResNet50 model (trained on ImageNet) is employed, with its final fully connected and softmax layers removed. Each input image is processed through the network, and the output from the global average pooling layer is extracted as a 2048-dimensional feature vector. This approach harnesses the powerful representational capacity of deep convolutional networks without necessitating end-to-end retraining.

B. SVM Classification

For classification, we adopt a hybrid approach by training a Support Vector Machine (SVM) on the features extracted from ResNet50. The SVM employs a radial basis function (RBF) kernel. Hyperparameters, including the regularization parameter C and the kernel parameter gamma, are optimized using grid search with cross-validation (GridSearchCV). This method is particularly effective when the available training data is limited, as it combines deep feature extraction with the robust classification capabilities of SVMs.

C. Training and Evaluation

The SVM classifier is trained on the extracted features from the training set. Model selection and hyperparameter tuning are performed using 5-fold cross-validation. The final model is evaluated on the validation set, and performance is assessed using accuracy, confusion matrix, and classification report metrics. Additionally, we visualize the training process by plotting accuracy and training time curves for different hyperparameter combinations.

V. EXPERIMENTS

A. Experimental Setup

All experiments were conducted on a workstation equipped with an NVIDIA GPU and PyTorch 2.0. Model performance was evaluated using accuracy, confusion matrix, and classification report (precision, recall, F1-score).

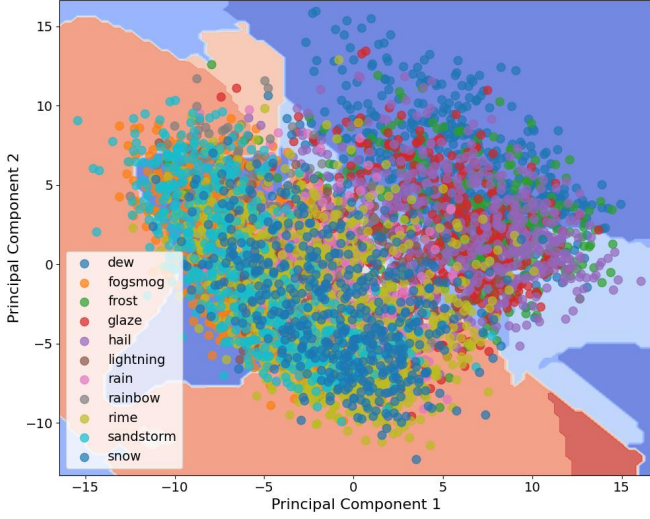


Fig. 2. Visualization of SVM decision boundaries and feature distribution after PCA dimensionality reduction.

B. Feature Visualization

To assess the discriminative power of the extracted features, we employed Principal Component Analysis (PCA) for visualization. Given the original 2048-dimensional feature space (output of ResNet50), PCA reduced the features to two dimensions. The x-axis ("Principal Component 1") and y-axis ("Principal Component 2") represent the two principal directions that retain the most variance from the original high-dimensional features.

As shown in Fig. 2, each point corresponds to a sample projected into this 2D space, and the background color indicates the SVM's decision regions for different weather categories. The clear clustering of samples from different classes demonstrates that the ResNet50 features are highly discriminative, and the SVM classifier is able to form effective decision boundaries even after dimensionality reduction. Overlapping or dispersed regions indicate categories that are more difficult to separate, which is consistent with the confusion matrix analysis.

C. SVM Training and Hyperparameter Tuning

The SVM classifier, equipped with an RBF kernel, was trained on the extracted features. Hyperparameters (C and gamma) were optimized via grid search with 5-fold cross-validation (GridSearchCV). Fig. 3 presents the training and validation accuracy, along with the fitting time for different hyperparameter combinations.

These results underscore the importance of hyperparameter tuning for achieving both high accuracy and computational efficiency in SVM-based classification. Specifically, grid search over the parameter space $\{C: [0.1, 1, 10, 100], \text{'gamma': ['scale', 'auto']}\}$ with 5-fold cross-validation (total 40 fits) identified the optimal SVM parameters as $C=10$ and $\text{gamma}=\text{'scale'}$, yielding a cross-validation accuracy of 0.8495. The total training time for SVM was approximately 1475 seconds.

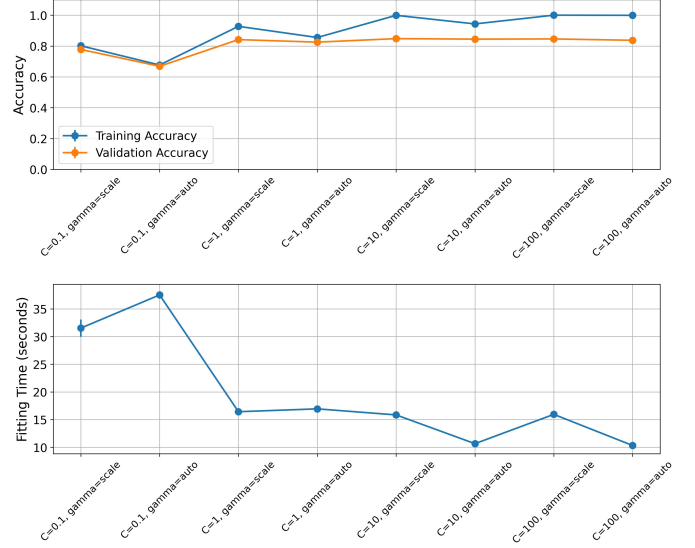


Fig. 3. Training and validation accuracy, and fitting time curves for different SVM hyperparameter combinations (C and gamma).

D. Classification Results and Confusion Matrix

The final ResNet50+SVM+PCA model was evaluated on the validation set. The confusion matrix, as shown in Fig. 4, illustrates the distribution of predictions across the weather classes. The model achieved high accuracy on most classes, with some confusion observed between visually similar categories such as rain and frost. Precision, recall, and F1-score for each class are summarized in Table II.

TABLE II. PERFORMANCE METRICS WEATHER CLASS

Class	precision	Recall	F1 Score
Dew	0.95	0.95	0.95
Fogsmog	0.87	0.87	0.87
Frost	0.72	0.81	0.76
Glaze	0.79	0.73	0.76
Hail	0.95	0.87	0.91
Lightning	0.99	1.00	0.99
Rain	0.86	0.90	0.88
Rainbow	1.00	0.85	0.92
Rime	0.83	0.87	0.85
Sandstorm	0.89	0.87	0.88
Snow	0.76	0.72	0.74

Categories such as lightning and dew achieve outstanding performance, with F1-scores of 0.99 and 0.95, respectively. In contrast, categories like frost and snow exhibit relatively lower F1-scores of 0.76 and 0.74. This discrepancy may be attributed to the high visual similarity between certain weather phenomena and the limited number of annotated samples in these categories.

The ResNet50+SVM+PCA model achieves high accuracy for most weather categories, with the majority of samples distributed along the diagonal of the confusion matrix. However, noticeable confusion exists between visually similar categories, such as frost and glaze, or rain and frost, where off-diagonal elements are more prominent. This confusion may be attributed to the high visual similarity and relatively small sample sizes. In contrast, categories like rime and lightning are

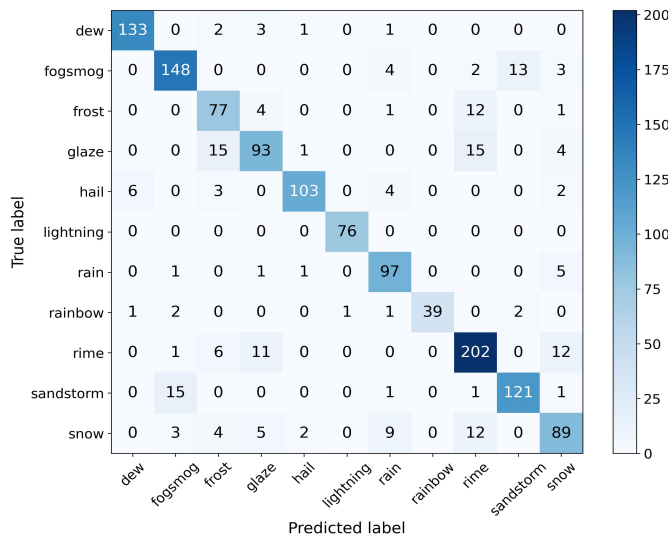


Fig. 4. ConConfusion matrix of the ResNet50+SVM model on the validation set.

classified with high precision, as evidenced by the strong diagonal elements and minimal misclassification. These results suggest that while the hybrid model is effective overall, further improvements could be made by addressing the misclassification of similar weather types, possibly through more advanced feature extraction or additional data augmentation.

VI. CONCLUSION

This study presented an effective approach for weather phenomenon classification by integrating deep feature extraction using ResNet50 with PCA feature extraction and SVM classification. This methodology addresses the challenges posed by limited training data through the strategic application of transfer learning and data augmentation techniques, including random cropping, horizontal flipping, and rotation.

Experimental results demonstrate that the proposed method achieves a validation accuracy of 85.80% across eleven weather categories. The model exhibits particularly strong performance on certain weather phenomena, such as lightning (F1-score: 0.99), dew (F1-score: 0.95), and rainbow (F1-score: 0.92), while achieving moderate results on more challenging categories like snow (F1-score: 0.72) and glaze (F1-score: 0.76).

Visualization of the feature space via PCA dimensionality reduction reveals clear clustering among different weather categories, confirming that ResNet50 is capable of extracting highly discriminative features from weather images. Furthermore, hyperparameter optimization of the SVM classifier enables a favorable balance between computational efficiency and classification performance.

Overall, this study demonstrates the effectiveness of hybrid approaches that combine deep feature extraction with traditional machine learning classifiers for weather phenomenon classification. The proposed methodology provides a practical solution for weather recognition systems, particularly in scenarios with limited annotated data.

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